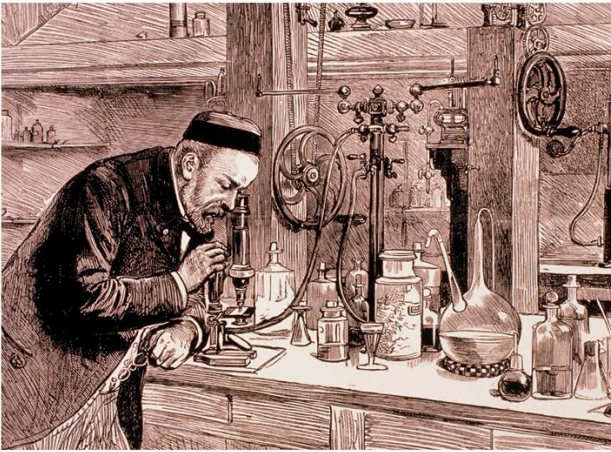
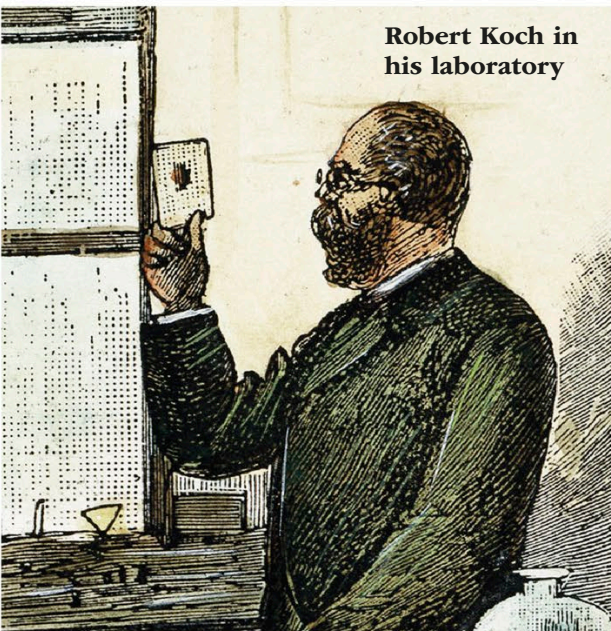


Louis Pasteur in his laboratory



GERM THEORY

The idea that infectious diseases are spread by germs, or microorganisms, is known as germ theory. Today, we take it for granted, but this view was highly controversial when Pasteur proposed it. With the help of the microscope, he showed that some tiny organisms were responsible for contaminating milk, while other microorganisms called yeasts caused the fermentation of beer and wine.

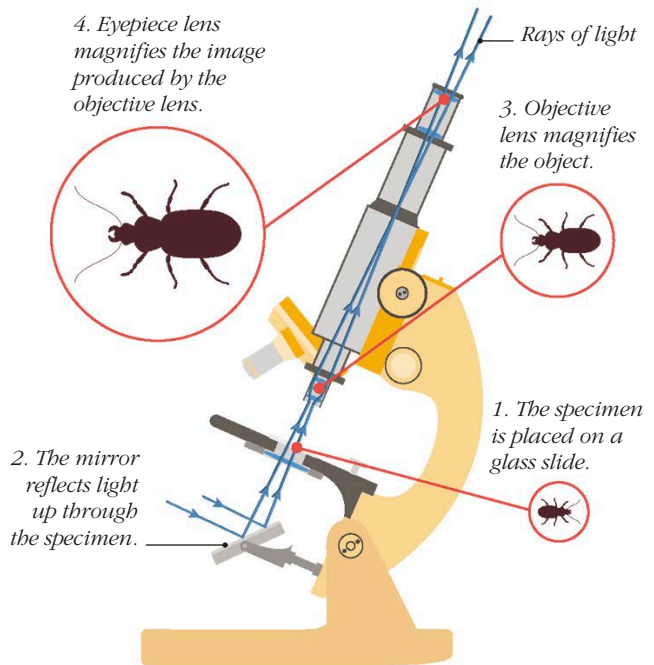


Robert Koch in his laboratory

FINDING BACTERIA

While Pasteur proved that diseases were caused by bacteria, the German scientist Robert Koch pinned down specific culprits. Microscopes enabled him to identify the bacteria that cause diseases such as anthrax (1876), tuberculosis (1882), and cholera (1883). For these discoveries, Koch is known as the “father of bacteriology” (the study of bacteria).

COMPOUND MICROSCOPE



A compound microscope operates in roughly the opposite way to a telescope. Rather than having a large lens for gathering faint light from far away, a microscope has a small lens for focusing light on small objects that are close up. Microscopes usually have a light source, an objective lens that can be focused, and a fixed eyepiece. The lenses focus by bending light to make the image appear larger.

ELECTRON MICROSCOPES

In the 1930s, the German physicist Ernst Ruska developed a microscope that could magnify up to 500,000 times.

Instead of using light, the device used a beam of electrons. It would eventually be possible to produce images of molecules and atoms. Today's most powerful microscopes can magnify up to 30 million times.

Image of bacteria (yellow) on the tip of a household pin, taken by an electron microscope

